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ORIGINAL ARTICLE

Scale-dependent turbulent coherent structures and pollutant fluxes over heated urban surfaces: spectral proper orthogonal decomposition and amplitude modulation analysis

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Abstract
Various scales of turbulent coherent structures (TCSs) individually and collectively contribute to pollutant dispersion in urban areas. Although it is well known that surface heating significantly modifies urban turbulent flows, its impact on TCSs with different scales and their interactions still needs further investigations. In this study, we examine the impact of surface heating on different scales of TCSs and their associated turbulent pollutant flux over an idealized building array. Two large-eddy simulations with and without surface heating are conducted using the PArallelized Large-eddy simulation Model (PALM). The spectral proper orthogonal decomposition (SPOD) and the amplitude modulation (AM) analysis are applied to extract different scales of TCSs and to examine their interactions, respectively. The surface heating enhances low-speed streaks at and above the roof level while it leads high-speed regions to be less organized into streaks. The surface heating preferentially enhances large-scale coherent structures and therefore substantially increases the relative importance of large-scale coherent structures for roof-level TKE and turbulent pollutant flux compared with mid-scale and small-scale coherent structures. The surface heating also enhances the modulation of small-scale turbulent momentum and pollutant fluxes by large-scale coherent structures above the roof level. This enhancement of the modulation mainly originates from the strengthened large-scale upward motions. This study emphasizes the necessity of applying modal decomposition and scale-interaction analysis together to better understand both scale-dependent characteristics of TCSs and interactions among TCSs with different scales.

Keywords Turbulent coherent structure · Pollutant flux · Urban surface heating · Large-eddy simulation · Spectral proper orthogonal decomposition · Amplitude modulation analysis

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1 Introduction

Rapid urbanization has led to increasingly dense building configurations and persistent air-quality concerns in many cities worldwide [1–4]. In urban areas, pollutants emitted from the surfaces are vertically transported across the building roof level and are effectively diluted in the overlying atmospheric boundary layer (ABL), the transport being dominated by turbulent flows rather than by mean flows [5–7]. In particular, turbulent coherent structures (TCSs) spanning a wide range of spatial and temporal scales exert a considerable influence on intermittent pollutant ejections and sweeps at the roof level, thus playing important roles in street-canyon ventilation [8–10].

TCSs in ABLs have been extensively studied through field observations, wind-tunnel experiments, and large-eddy simulations (LESs) [8, 11–15]. In ABLs, it is known that elongated motions such as large-scale motions and very large-scale motions carry substantial fractions of turbulent kinetic energy (TKE) and shear stress and that hairpin packets and shear-layer vortices contribute to mixing therein [16–18]. It is also known that vortices induced by shear-layer instabilities at the roof level lead to ejections and sweeps that dominate pollutant removal from street canyons [7, 8, 10].

Recent LES studies have attempted to disentangle TCSs with different scales and/or to examine their contributions to the ventilation through modal decomposition methods [10, 19–22]. Shah and Bou-Zeid [20] examined the dynamics of very large-scale motions using the proper orthogonal decomposition (POD) and found that small-scale motions are embedded in large-scale streamwise rolls and these rolls are dominant under most stability conditions. As an extension of the POD, the spectral POD (SPOD) considers both spatial pattern and frequency of turbulent flows in decomposing them into distinct modes, which can better capture the characteristics of different scales of TCSs compared with the POD that only considers the spatial pattern [18, 23]. Using the SPOD in a two-dimensional street canyon under a neutral condition, Zhang et al. [10, 19] showed that external large-scale structures carry a substantial fraction of TKE and that smaller Kelvin–Helmholtz vortices in the roof-level shear layer contribute more directly to pollutant removal. They also highlighted that the modes with the largest fraction of TKE do not necessarily dominate pollutant fluxes at the roof level. This result underscores the need for scale-resolved examinations of TCSs for in-depth understanding of pollutant dispersion in urban areas.

The SPOD-based studies of street-canyon ventilation have considered a neutral condition [10, 19]. The investigation of the impact of surface heating on the multiscale organization of coherent motions and their contributions to pollutant exchanges across the roof level, utilizing the advantage of the SPOD, will be beneficial for improving the understanding of pollutant dispersion in urban areas but has not been conducted yet. This motivates the present study. While the SPOD enables identifying the individual characteristics of TCSs with different scales, it is limited in examining the interactions among TCSs across scales, which also importantly contribute to the pollutant dispersion [22]. The amplitude modulation (AM) analysis is a useful tool to disentangle these interactions, making it possible to investigate how buoyancy-enhanced large-scale motions influence small-scale turbulent pollutant exchanges across the roof level [22, 24–26]. This also motivates the present study.

In this study, we examine how the surface heating changes different scales of TCSs and their roles in pollutant dispersion over urban surfaces, particularly at the roof level, through

the SPOD and AM analysis. For this, LESs are conducted over an idealized building array with and without surface heating.

2 Methodology

2.1 Numerical model and simulation design

The large-eddy simulations are carried out using the PArallelized Large-eddy simulation Model (PALM), version 6.0 [27], which is widely used for simulations of urban turbulent flows and pollutant dispersion. PALM solves the Navier–Stokes equations in the Boussinesq approximation. Subgrid-scale turbulence is parameterized using the 1.5-order Deardorff turbulence closure [28, 29]. For spatial and temporal discretizations, the model employs the fifth-order Wicker–Skamarock scheme on the Arakawa C-grid and third-order Runge–Kutta scheme, respectively. Further information about PALM is provided in Maronga et al. [27].

Figure 1 illustrates an array of 10 × 10 identical buildings adopted in the present study. All buildings are cubic, and their heights (H) are 32 m. The distance between neighboring buildings in both streamwise and spanwise directions is also 32 m. The computational domain spans 640 m × 640 m in the horizontal directions and 1508 m in the vertical direction. The grid size in the horizontal directions is uniformly 2 m. The grid size in the vertical direction is 2 m below $z = 200$ m, gradually stretched with a ratio of 1.08 up to $z = 303$ m, and 10 m above $z = 303$ m. The cyclic boundary condition is applied to velocity, potential temperature, and pollutant concentration in the streamwise and spanwise directions. At the bottom boundary, the no-slip boundary condition is imposed for velocity and the surface momentum and heat fluxes are diagnosed using the Monin–Obukhov similarity with a roughness length of 0.01 m. At the top boundary, the Neumann condition is applied. The initial wind is aligned with the streamwise direction with a uniform speed (U_{ref}) of 8 m s^{-1}. The initial potential temperature and pollutant concentration are set to 300 K and 0 μg m^{-3}, respectively, throughout the computational domain. The pollutant is released with a uniform flux of 1 μg m^{-2} s^{-1} at the road surfaces. A simulation with a uniform surface heat flux of 0.3 K m s^{-1} on the building roof, wall, and road surfaces (heating simulation) is conducted as well as a simulation without surface heat flux (no-heating simulation). The numerical

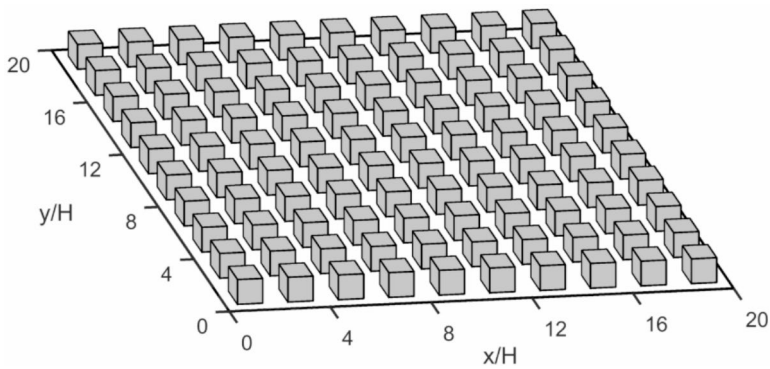

Fig. 1 Illustration of the building array. H denotes the building height

model is integrated for a total of 3 h, and the last 1 h is considered as the analysis period, during which the streamwise flow and turbulent fluxes are in a quasi-steady state.

2.2 Spectral proper orthogonal decomposition

To extract scale-dependent TCSs and quantify their contributions to pollutant transport, the SPOD is applied to the simulation data. The SPOD is a frequency-domain extension of the POD in which the inner product is formulated in a space–time sense [23, 30]. Based on the assumption of statistical stationarity in turbulence over the analysis period, the SPOD yields a set of modes that are optimally energetic and mutually orthogonal at each frequency and that remain coherent in both space and time.

Let $\mathbf{u}(\mathbf{x}, t)$ denote the velocity vector in the sampling domain, where $\mathbf{x}$ is the spatial coordinate and t is the time. The fluctuating component $\mathbf{u}'(\mathbf{x}, t)$ is obtained by subtracting the temporal average from $\mathbf{u}(\mathbf{x}, t)$. The prime ($'$) is used to denote the deviation from the temporal average. Assuming statistical stationarity over the analysis period, $\mathbf{u}'$ can be represented as a continuous superposition of harmonic contributions $\hat{\mathbf{u}}(\mathbf{x}, f)\, e^{i2\pi ft}$ at different frequencies:

$$\mathbf{u}'(\mathbf{x}, t) = \int_{-\infty}^{+\infty} \hat{\mathbf{u}}(\mathbf{x}, f)\, e^{i2\pi ft} df, \tag{1}$$

$$\hat{\mathbf{u}}(\mathbf{x}, f) = \sum_{n} a_{n,f} \mathbf{\Psi}_{n,f}(\mathbf{x}), \tag{2}$$

where $\hat{\mathbf{u}}$ is the temporal Fourier transform of $\mathbf{u}'$ at frequency f, $a_{n,f}$ is the n-th mode coefficient at frequency f, and $\mathbf{\Psi}_{n,f}(\mathbf{x})$ is the n-th spatial mode at frequency f. Equation (2) states that the Fourier coefficient field at each frequency is expressed as a linear combination of spatial basis functions $\mathbf{\Psi}_{n,f}(\mathbf{x})$ with their amplitudes $a_{n,f}$. The frequency f can be regarded as the temporal scale of the corresponding harmonic contribution comprising $\mathbf{u}'$. The spatial distribution of a mode and corresponding coefficient at low (high) frequencies can reflect the pattern and intensity of large-scale (small-scale) coherent structures.

The SPOD modes and coefficients are uniquely determined by imposing orthogonality conditions. The SPOD modes are orthonormal with respect to a volume-averaged inner product as

$$\frac{1}{V} \int \mathbf{\Psi}_{m,f}^{H}(\mathbf{x}) \mathbf{\Psi}_{n,f}(\mathbf{x}) dV = \delta_{nm}, \tag{3}$$

where V is the volume of the sampling domain, $\mathbf{\Psi}_{m,f}^{H}(\mathbf{x})$ is the Hermitian transpose of $\mathbf{\Psi}_{m,f}(\mathbf{x})$, and δ_{nm} is the Kronecker delta (1 if $n=m$ and 0 if $n \neq m$). The SPOD mode coefficients also satisfy the following orthogonality:

$$\frac{1}{T} a_{n,f} a_{m,f}^{*} = \lambda_{n,f} \delta_{nm}, \tag{4}$$

where T is the temporal length, $a^*_{m,f}$ is the complex conjugate of $a_{m,f}$, and $\lambda_{n,f}$ is the mode energy associated with $\mathbf{\Psi}_{n,f}(\mathbf{x})$. By summing $\lambda_{n,f}$ across all modes and integrating over all frequencies, the domain-averaged TKE is recovered as

$$2 \times \mathrm{TKE} = \int_{-\infty}^{+\infty} \sum_n \lambda_{n,f}\, df. \tag{5}$$

Equations (3)–(5) mean that $\lambda_{n,f}$ measures the contribution of n-th mode at frequency f to the TKE. Therefore, a set of mode energies $\lambda_{n,f}$ provides the respective contributions of scale-resolved coherent structures to the TKE.

In practice, we compute the SPOD following the snapshot-based algorithm of Towne et al. [30], which combines the snapshot POD method with the Welch's method [31]. The total time window of $\mathbf{u}(\mathbf{x}, t)$ is divided into overlapping blocks. A tapering window is applied to each block to reduce the spectral leakage, and the discrete Fourier transform is performed in time. At each discrete frequency, the cross-spectral density tensor is formed from the blockwise Fourier coefficients and an eigenvalue problem is solved to obtain the leading eigenvalues $\lambda_{n,f}$ and eigenfunctions $\mathbf{\Psi}_{n,f}(\mathbf{x})$. Repeating this procedure over all discrete frequencies yields a finite set of SPOD modes and mode energies that capture the scale-dependent coherent structures and their energy contents, respectively.

To quantify the contributions of the SPOD modes to vertical turbulent pollutant fluxes, we calculate the SPOD cospectrum introduced by Zhang et al. [10]. The vertical velocity deviation $w'(\mathbf{x}, t)$ and pollutant concentration deviation $c'(\mathbf{x}, t)$ are considered by subtracting their respective temporal averages. The temporal Fourier transforms of $w'(\mathbf{x}, t)$ and $c'(\mathbf{x}, t)$ are denoted by $\hat{w}(\mathbf{x}, f)$ and $\hat{c}(\mathbf{x}, f)$, respectively. For the n-th mode at frequency f, the SPOD cospectrum between vertical velocity and pollutant concentration is defined following Zhang et al. [10]:

$$S^{(wc)}_{n,f}(\mathbf{x}) = \mathrm{Re}\left(\frac{1}{T} a_{n,f} \Psi^{(w)}_{n,f}(\mathbf{x}) \hat{c}^*(\mathbf{x}, f)\right), \tag{6}$$

where $\mathrm{Re}(\cdot)$ is the real part of a complex quantity, $\Psi^{(w)}_{n,f}(\mathbf{x})$ is the vertical component of $\mathbf{\Psi}_{n,f}(\mathbf{x})$, and $\hat{c}^*(\mathbf{x}, f)$ is the complex conjugate of $\hat{c}(\mathbf{x}, f)$. In Eq. (6), $S^{(wc)}_{n,f}(\mathbf{x})$ represents the contribution of n-th mode at frequency f to the vertical turbulent pollutant flux. The vertical turbulent pollutant flux is recovered by summing over all modes and integrating over all frequencies as

$$\overline{w'(\mathbf{x}, t) c'(\mathbf{x}, t)} = \int_{-\infty}^{+\infty} \sum_n S^{(wc)}_{n,f}(\mathbf{x}) df, \tag{7}$$

where the overbar ($\bar{\ }$) represents the averaging over time. Here, the flux contributions from the correlations of the harmonic contributions of $w'(\mathbf{x}, t)$ and $c'(\mathbf{x}, t)$ at different frequencies are not included, as they are considered to be averaged out by the temporal average. This SPOD cospectrum provides a scale-resolved decomposition of the vertical turbulent pollutant flux. In estimating the SPOD cospectrum, we employ the Welch's method to both velocity and pollutant concentration following Zhang et al. [10].

2.3 Amplitude modulation analysis

To quantitatively examine the interactions between large-scale and small-scale turbulences, the AM analysis is conducted following Liu et al. [22] and Zhou et al. [26]. First, the turbulent fluctuations are decomposed into frequency-band-limited components based on frequency filtering. For given variables $a'(\mathbf{x}, t)$ and $b'(\mathbf{x}, t)$, the low-frequency band (large-scale) component $a_{\mathrm{L}}(\mathbf{x}, t)$ and high-frequency band (small-scale) component $b_{\mathrm{H}}(\mathbf{x}, t)$ are extracted by applying band-pass filters in the frequency domain, respectively. The instantaneous amplitude of the small-scale component is calculated based on its envelope. The degree of amplitude modulation of $b_{\mathrm{H}}(\mathbf{x}, t)$ by $a_{\mathrm{L}}(\mathbf{x}, t)$ is expressed as the correlation coefficient:

$$R_{a_{\mathrm{L}},b_{\mathrm{H}}} = \frac{\overline{a_{\mathrm{L}} E_{\mathrm{L}}(b_{\mathrm{H}})}}{\sqrt{\overline{a_{\mathrm{L}}{}^2}}\sqrt{\overline{E_{\mathrm{L}}{}^2(b_{\mathrm{H}})}}}, \tag{8}$$

where $E_{\mathrm{L}}(\cdot)$ denotes the large-scale envelope. A positive (negative) AM coefficient means that large-scale fluctuations are positively (negatively) correlated with the amplitude of small-scale fluctuations.

3 Results and discussion

3.1 Vertical profiles of mean and turbulent quantities

Figure 2 shows horizontally and temporally averaged vertical profiles of the mean and turbulent quantities during t = 2–2.5 h in the heating and no-heating simulations. Here, the angle brackets ($\langle\ \rangle$) denote the horizontal averaging at a given height [32, 33]. The streamwise velocity is lower in the heating simulation than in the no-heating simulation (Fig. 2a). In both simulations, the pollutant concentration decreases with height (Fig. 2b). The pollutant concentration in the heating simulation is notably lower than that in the no-heating simulation particularly near the ground. This originates from the enhanced vertical turbulent mixing due to the surface heating. The mean pollutant concentration below the roof level is 7.07 μg m^{-3} in the heating simulation and 26.23 μg m^{-3} in the no-heating simulation. In the heating simulation, due to the surface heating, the potential temperature decreases with height and the layers near the ground and near the roof level are prominently unstable (Fig. 2c).

The TKE in the heating simulation is much larger than that in the no-heating simulation at all levels (Fig. 2d). Below the roof level, the peak value of TKE is found at z/H=0.3750 (0.6875) in the heating (no-heating) simulation, being 2.80 (0.81) m^2 s^{-2}. While the vertical velocity variance is very small in the no-heating simulation, it is appreciable in the heating simulation with a distinct peak of 1.54 m^2 s^{-2} at z/H=0.6875 (Fig. 2e), contributing to the larger TKE in the heating simulation (Fig. 2d). Above the roof level, the vertical velocity variance in the heating simulation monotonically increases with height. In both simulations, the vertical turbulent momentum flux is negative, meaning the downward turbulent transport of momentum (Fig. 2f). The downward turbulent momentum flux is consider-

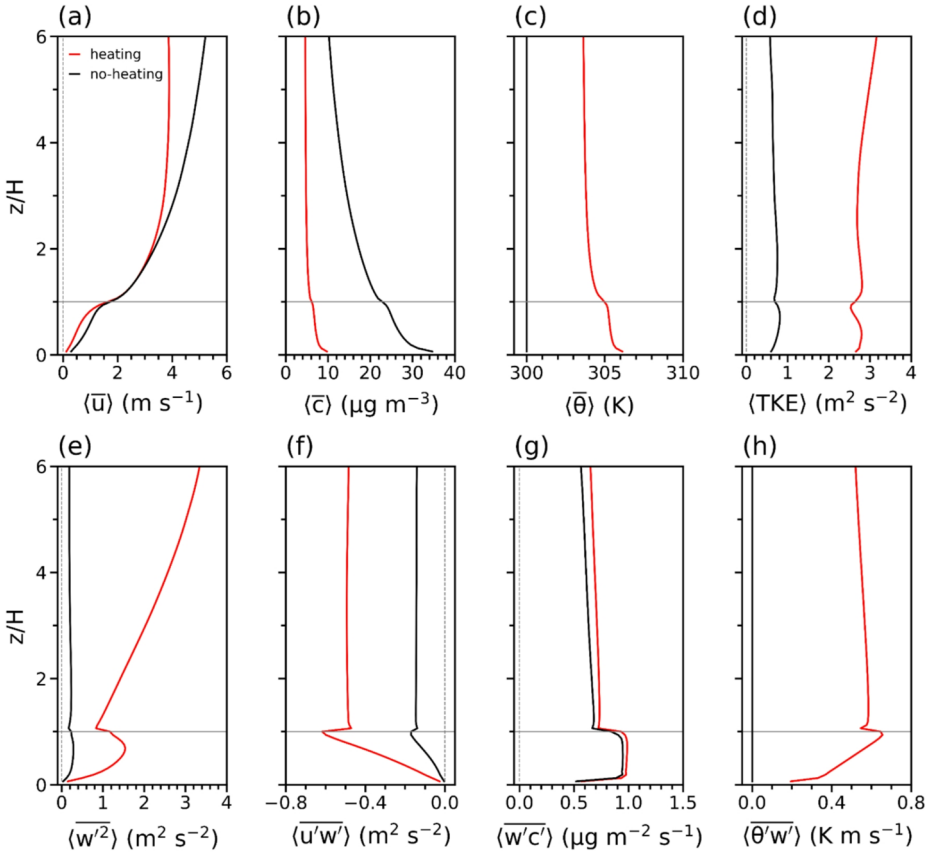

Fig. 2 Horizontally and temporally averaged vertical profiles of **a** streamwise velocity, **b** pollutant concentration, **c** potential temperature, **d** turbulent kinetic energy, **e** vertical velocity variance, **f** vertical turbulent momentum flux, **g** vertical turbulent pollutant flux, and **h** vertical turbulent heat flux during t=2–2.5 h in the heating (red) and no-heating (black) simulations. The turbulent quantities in (**d**–**h**) are resolved ones

ably enhanced in the heating simulation compared to the no-heating simulation, maximized at the roof level in both simulations. The vertical turbulent pollutant flux exhibits similar variations with height in the heating and no-heating simulations, and it is slightly larger in the heating simulation than in the no-heating simulation (Fig. 2g). The maximum vertical turbulent pollutant flux in the heating (no-heating) simulation is 0.99 (0.95) μg m^{-2} s^{-1} at z/H=0.7500 (0.5625). In the heating simulation, the vertical turbulent heat flux is positive at all levels, meaning the upward turbulent transport of heat (Fig. 2h). The maximum upward turbulent heat flux of 0.66 K m s^{-1} appears near the roof level in the heating simulation.

3.2 Turbulent flow and pollutant concentration fields

The instantaneous horizontal distributions of streamwise velocity deviation at the roof level and at the upper level (z/H = 2) in the heating and no-heating simulations are shown in Fig. 3. The strong updraft and downdraft regions are indicated by the yellow-green and

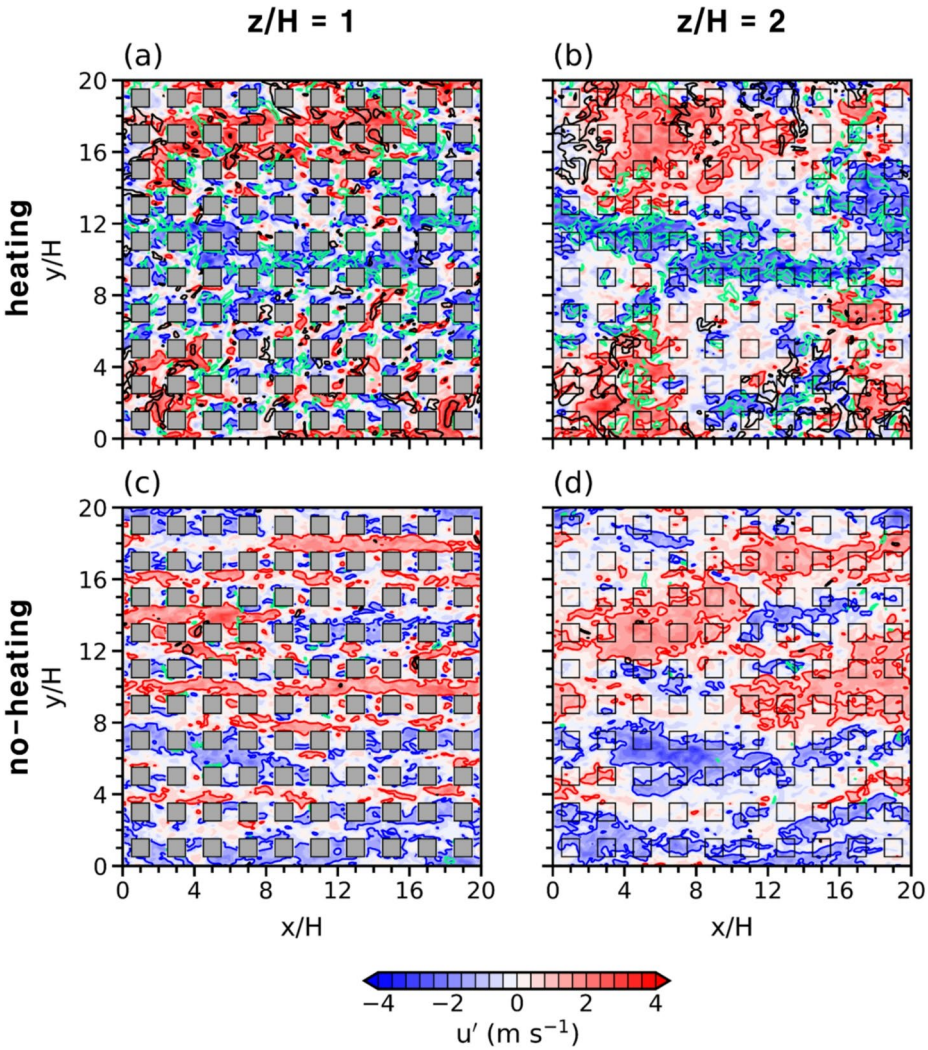

Fig. 3 Fields of streamwise velocity deviation (shaded) at z/H = **a**, **c** 1 and **b**, **d** 2 at t=2 h in the **a**, **b** heating and **c**, **d** no-heating simulations. The standardized streamwise velocity deviations of 1 and −1 are indicated by the red and blue contours, respectively. The yellow-green and black contours indicate vertical velocity deviations of 1.5 m s^{-1} and −1.5 m s^{-1}, respectively

black contours, respectively. In the no-heating simulation, elongated low-speed and high-speed streaks are aligned in the streamwise direction at both levels (Fig. 3c, d). The strong updraft and downdraft regions are rarely found. Compared with the no-heating simulation, in the heating simulation, elongated low-speed streaks are more pronounced at both levels while high-speed regions are less organized into streamwise elongated streaks (Fig. 3a, b). The pronounced low-speed streaks could be associated with the enhanced updrafts due to the surface heating. Both low-speed and high-speed regions exhibit stronger correlations between the roof level and upper level in the heating simulation than in the no-heating

simulation. The strong updraft and downdraft regions are clearly seen at both levels in the heating simulation and tend to coincide with the low-speed streaks and high-speed regions, respectively. This leads to the enhanced downward turbulent momentum flux in the heating simulation compared with the no-heating simulation (Fig. 2f). The surface heating overall enhances the intensities of low-speed and high-speed structures, as well as updrafts and downdrafts. Park and Baik [34] also found vertically well-developed streamwise-elongated low-speed regions and the enhancement of updrafts above the roof level in the presence of bottom heating.

Figure 4 presents the instantaneous horizontal distributions of pollutant concentration at the roof level in the heating and no-heating simulations. In both simulations, the relatively high and low pollutant concentration regions generally coincide with the low-speed and high-speed regions, respectively, as found in previous LES studies [5, 8]. In the no-heating simulation, the pollutant removal through the roof level by vertical turbulent flows is relatively inefficient as the strong updrafts and downdrafts are nearly absent (Fig. 4b). On the other hand, in the heating simulation, due to the pronounced strong updrafts and downdrafts originating from the surface heating, the pollutants are efficiently removed through the roof level (Fig. 4a). The pollutant concentration at the roof level is clearly lower in the heating simulation than in the no-heating simulation, the mean value being 5.94 (21.80) μg m^{-3} in the heating (no-heating) simulation.

3.3 Scale-dependent characteristics of TCSs and their roles in pollutant flux

In this subsection, the scale-dependent characteristics of coherent structures and their roles in vertical turbulent pollutant flux are compared between the heating and no-heating simula-

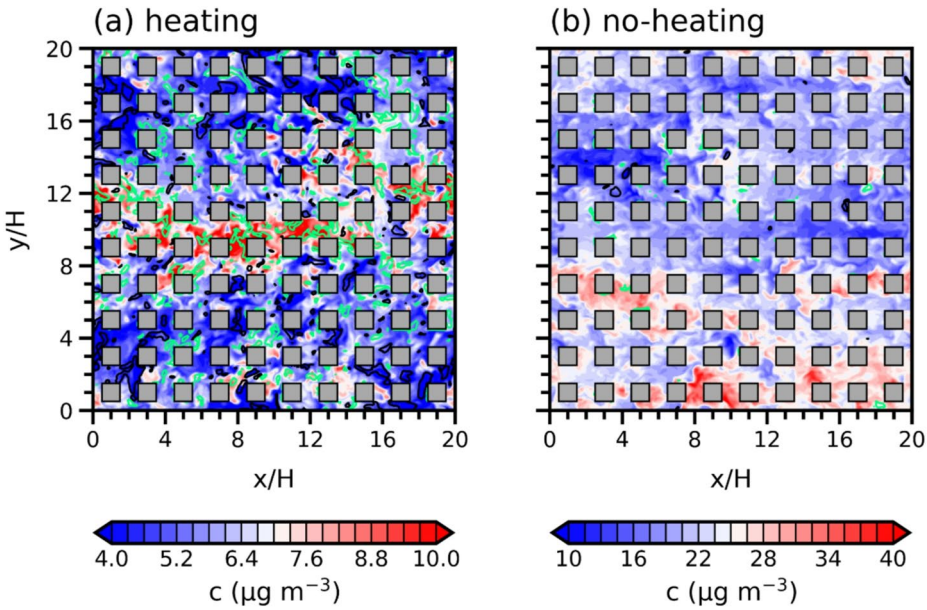

Fig. 4 Fields of pollutant concentration (shaded) at $z/H=1$ at $t=2$ h in the **a** heating and **b** no-heating simulations. The yellow-green and black contours indicate vertical velocities of 1.5 m s^{-1} and −1.5 m s^{-1}, respectively

tions using the SPOD. The SPOD is applied to streamwise velocity, vertical velocity, and pollutant concentration at $x/H = 9$ which is the closest y-z plane to the domain center among those crossing the building centers. For the SPOD, the sampling interval is set to 1 s and a total of five blocks whose temporal length is 1200 s are considered. The overlapping rate between blocks is set to 50%. To reduce the spectral leakage, the discrete Fourier transform is conducted after applying the Hamming window function to the blocks [30]. Note that the domain-averaged TKE is totally explained by the decomposed five modes.

The SPOD spectra at $x/H=9$ at the roof level during $t=2$–3 h in the heating and no-heating simulations are presented in Fig. 5. The abscissa is the non-dimensional frequency fH/U_{ref}, and the ordinate is the non-dimensional mode energy obtained through normalizing each $\lambda_{n,f}$ by the domain-averaged TKE. In both simulations, the first mode exhibits much larger energies than the higher modes at almost all frequencies (Fig. 5). The leading SPOD mode captures a substantial fraction of the total TKE in both heating (40.1%) and no-heating (29.6%) simulations. The mode energy decreases from the second to fifth modes, indicating the smaller contribution to the TKE with increasing mode number. In the no-heating simulation, the mode energy peaks at $fH/U_{\mathrm{ref}} = 0.030$ in the first mode and the mode energy tends to decrease as the frequency increases (Fig. 5b). Compared with the no-heating simulation, the mode energy in the heating simulation exhibits a much larger peak at $fH/U_{\mathrm{ref}} = 0.023$ and also tends to decrease with increasing frequency (Fig. 5a). The prominently larger peak mode energy at relatively low frequency in the heating simulation is attributed to the surface heating. To quantitatively compare the contributions of different scales of coherent structures to the TKE between the heating and no-heating simulations, the low-frequency band (L band), middle-frequency band (M band), and high-frequency band (H band) are defined as $fH/U_{\mathrm{ref}} = 0.013$–0.043, 0.063–0.093, and 0.140–0.170, respectively. The frequency ranges for the L, M, and H bands are selected based on the SPOD spectra and the SPOD cospectra between vertical velocity and pollutant concentration, exhibiting large mode energies and/or appreciable contributions to the roof-level vertical turbulent pollutant flux. Note that the widths of L, M, and H bands are the same as each other.

Table 1 shows the contributions of L, M, and H bands to the TKE at the roof level for the first mode and all modes in the heating and no-heating simulations. The TKE contribu-

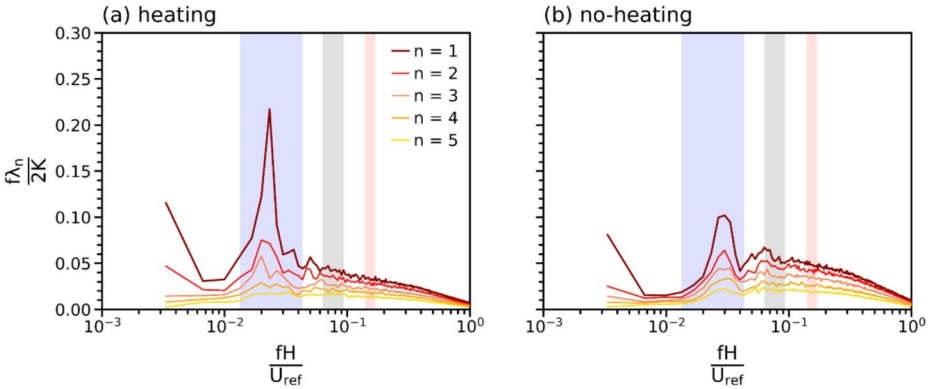

Fig. 5 Spectral proper orthogonal decomposition spectra at $x/H=9$ and at $z/H=1$ during $t=2$–3 h in the **a** heating and **b** no-heating simulations. The solid lines with different colors indicate the spectra of the first to fifth modes. The blue, gray, and red shaded areas indicate the low-frequency band (L band), middle-frequency band (M band), and high-frequency band (H band), respectively

Table 1 Band-integrated contributions of the mode energy to the TKE for the first mode and all modes at $x/H=9$ and $z/H=1$ in the heating and no-heating simulations

	Heating			No-heating		
	L band	M band	H band	L band	M band	H band
$\int_{\mathrm{B}} \lambda_{n=1,f}\ df$	0.47 (45.5%)	0.06 (5.6%)	0.02 (2.4%)	0.04 (23.3%)	0.02 (9.9%)	0.01 (4.3%)
$\int_{\mathrm{B}} \sum_{n} \lambda_{n,f}\ df$	0.98 (38.1%)	0.20 (7.6%)	0.08 (3.2%)	0.14 (25.5%)	0.06 (10.9%)	0.03 (4.8%)

The contribution of each band is given in the unit of $\mathrm{m}^2\ \mathrm{s}^{-2}$

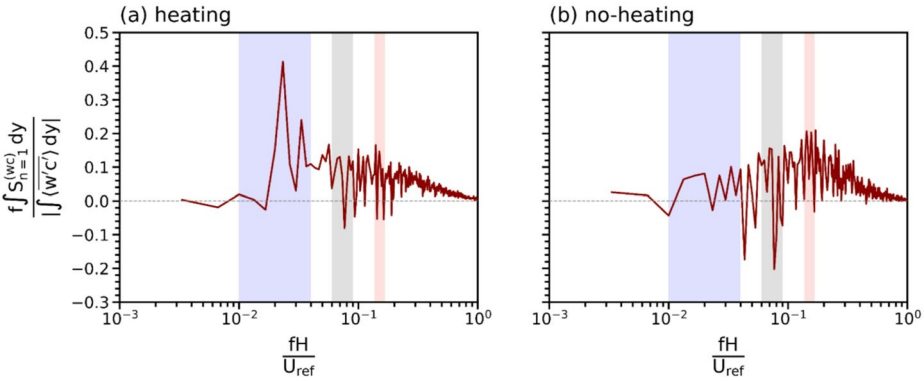

Fig. 6 Spectral proper orthogonal decomposition cospectra between vertical velocity and pollutant concentration for the first mode at $x/H=9$ and at $z/H=1$ during $t=2$–3 h in the **a** heating and **b** no-heating simulations. The blue, gray, and red shaded areas indicate the low-frequency band (L band), middle-frequency band (M band), and high-frequency band (H band), respectively

tions of all three bands are larger in the heating simulation than in the no-heating simulation (Table 1) as the TKE increases due to the surface heating (Fig. 2d). The increase in the TKE contributions due to the surface heating is more pronounced particularly in the L band than in the M and H bands. For the first mode, the contribution of L band increases from 0.04 to 0.47 $\mathrm{m}^2\ \mathrm{s}^{-2}$. This leads to a substantial increase in the relative importance of the L band from 23.3 to 45.5%. The contribution of M (H) band also increases from 0.02 (0.01) to 0.06 (0.02) $\mathrm{m}^2\ \mathrm{s}^{-2}$, but its relative importance decreases. For all modes, the changes in the contributions and relative importance of the three bands are qualitatively consistent with those for the first mode. The results of Fig. 5 and Table 1 show that the surface heating causes the roof-level TKE to be more dominated by low-frequency (large-scale) coherent structures. These results are in accordance with Shah and Bou-Zeid [20] which reveals through the POD that the surface buoyancy flux significantly strengthens large-scale streamwise rolls.

Figure 6 shows the SPOD cospectra between vertical velocity and pollutant concentration for the first mode at $x/H = 9$ and at the roof level during $t = 2$–3 h in the heating and no-heating simulations. The cospectra are normalized by the magnitude of temporally and horizontally averaged vertical turbulent pollutant flux at the roof level. In both simulations, the SPOD cospectrum is predominantly positive across frequencies, indicating that the first mode is primarily associated with upward turbulent pollutant flux at the roof level (Fig. 6). In the no-heating simulation, the SPOD cospectrum tends to increase with increasing

frequency in the frequency range encompassing the L to H bands (Fig. 6b), which contrasts with the SPOD spectrum (Fig. 5b). Zhang et al. [10] also found a similar increasing tendency of the SPOD cospectrum with increasing frequency under a neutral condition. Meanwhile, the SPOD cospectrum in the heating simulation exhibits distinct positive peaks (at $fH/U_{\mathrm{ref}} = 0.023$ and 0.033) in the L band and tends to decrease with increasing frequency in the frequency range encompassing the L to H bands (Fig. 6a). This reveals that the surface heating causes low-frequency (large-scale) coherent structures to be more important in removing pollutants through the roof level.

The contributions of the L, M, and H bands to vertical turbulent pollutant flux at $z/H = 0.0625$ and the roof level for the first mode and all modes in the heating and no-heating simulations are shown in Table 2. The $z/H = 0.0625$ level corresponds to $z = 2$ m, near the pedestrian level. At the roof level, the contribution of the L band is the largest among those of the three bands, accounting for 42.1% (23.9%) for the first mode and 27.0% (15.1%) for all modes in the heating (no-heating) simulation (Table 2). It is also noticeable that the contribution of the L band to vertical turbulent pollutant flux in the heating simulation is much larger than that in the no-heating simulation (Table 2), which is similar to the change in the contribution of the L band to the TKE due to the surface heating (Table 1). This indicates that the surface heating leads to more effective pollutant removal through the roof level mainly by enhancing low-frequency (large-scale) coherent structures. Near the pedestrian level, the L band exhibits the largest contribution to vertical turbulent pollutant flux among all three bands as well, accounting for 22.3% (26.2%) for the first mode and 16.6% (19.5%) for all modes in the heating (no-heating) simulation. Different from the roof level, near the pedestrian level close to the pollutant emission sources, the contribution of the L band to vertical turbulent pollutant flux does not largely change due to the surface heating.

To examine how pollutants are dispersed by different scales of coherent structures in and above street canyons, the spatial distributions of upward turbulent pollutant flux contributed by each of the three bands for the first mode are analyzed. For each of the L, M, and H bands, the band-integrated SPOD cospectrum between vertical velocity and pollutant concentration for the first mode $S_{\mathrm{B},1}^{(wc)}$ is calculated as

$$S_{\mathrm{B},1}^{(wc)} = \int_{\mathrm{B}} S_{n=1,f}^{(wc)}(\mathbf{x})\ df. \tag{9}$$

Table 2 Band-integrated contributions of the spectral proper orthogonal decomposition cospectrum between vertical velocity and pollutant concentration to vertical turbulent pollutant flux for the first spatial mode and all spatial modes at $x/H = 9$, and at $z/H = 0.0625$ and $z/H = 1$ in the heating and no-heating simulations

		Heating			No-heating		
		L band	M band	H band	L band	M band	H band
$\int_{\mathrm{B}} S_{n=1,f}^{(wc)}(\mathbf{x})df$	$z/H = 0.0625$	0.015 (22.3%)	0.007 (11.1%)	0.003 (4.9%)	0.017 (26.2%)	0.005 (7.2%)	0.006 (9.0%)
	$z/H = 1$	0.063 (42.1%)	0.014 (9.2%)	0.006 (4.2%)	0.025 (23.9%)	0.006 (5.6%)	0.011 (9.9%)
$\int_{\mathrm{B}} \sum_{n} S_{n,f}^{(wc)}(\mathbf{x})df$	$z/H = 0.0625$	0.041 (16.6%)	0.027 (10.9%)	0.017 (7.0%)	0.046 (19.5%)	0.036 (15.0%)	0.021 (8.9%)
	$z/H = 1$	0.122 (27.0%)	0.049 (10.9%)	0.028 (6.3%)	0.066 (15.1%)	0.050 (11.5%)	0.042 (9.7%)

The contribution of each band is given in the unit of $\mu g\ m^{-2}\ s^{-1}$

Figure 7 presents the spatial distributions of $\mathrm{sgn}(w'_{\mathrm{B},1}) \times \max(0, S^{(wc)}_{\mathrm{B},1})$ at $x/H=9$ during $t=2$–3 h in the heating simulation. The positive (negative) areas correspond to the regions where the first mode produces upward turbulent pollutant flux associated with the updrafts (downdrafts), namely pollutant ejection (sweep). In the L band, large-scale vortices result in the vertically well-organized areas of prominent upward turbulent pollutant flux near and above the roof level (Fig. 7a). The well-organized areas of pollutant ejection and sweep are found at $y/H \sim 4.5$–6 and 6–11, respectively. In the M band, mid-scale vortices play important roles in upward turbulent pollutant flux (Fig. 7b), whose areas are less organized than those resulting from the large-scale vortices in the L band. In the H band, the upward turbulent pollutant flux is mainly driven by small-scale irregular vortices (Fig. 7c). The areas of pollutant ejection and sweep developed by the small-scale vortices are more prominent near the roof level than well above the roof level.

Figure 8 is the same as Fig. 7 but for the no-heating simulation. In the no-heating simulation, the scales of vortices in the L band are overall smaller than those in the heating simulation (Fig. 8a). Also, the upward turbulent pollutant flux resulting from the vortices in the L

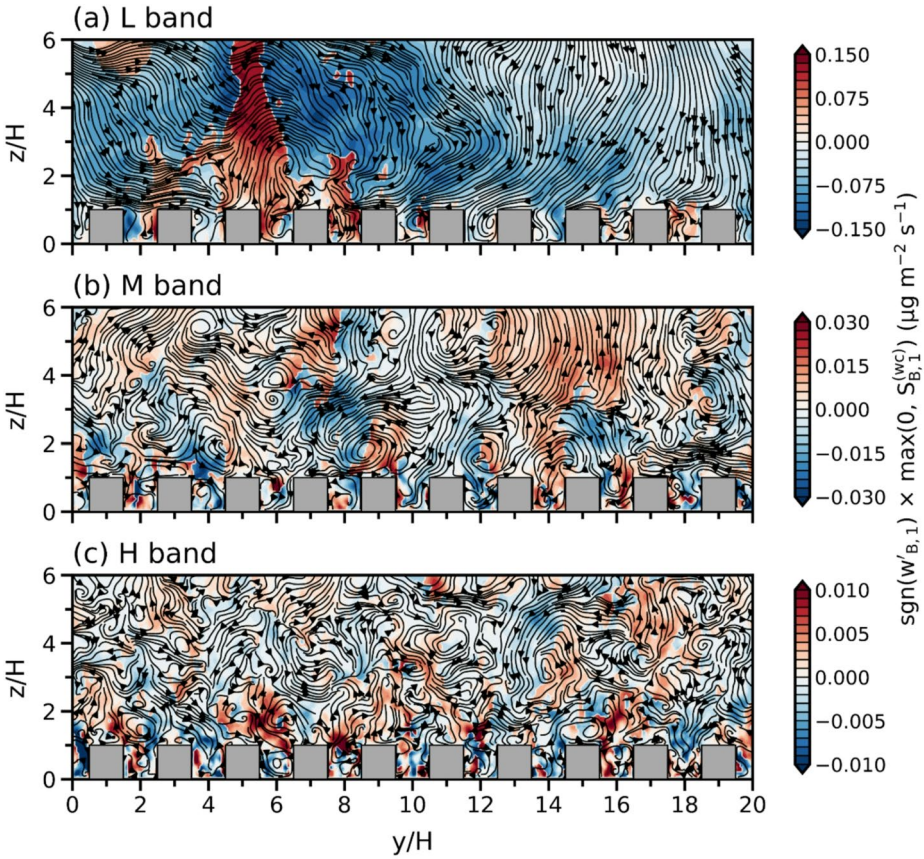

Fig. 7 Fields of $\mathrm{sgn}(w'_{\mathrm{B},1}) \times \max(0, S^{(wc)}_{\mathrm{B},1})$ (shaded) for the L, M, and H bands at $x/H=9$ during $t=2$–3 h in the heating simulation. Streamlines of the spanwise and vertical components of the first mode for each band are also shown

band in the no-heating simulation is much less pronounced than that in the heating simulation. In the M band, mid-scale vortices importantly contribute to both pollutant ejection and sweep (Fig. 8b), with overall scales of vortices and magnitudes of upward turbulent pollutant flux being similar to those in the heating simulation. Compared with the heating simulation, the small-scale vortices in the H band in the no-heating simulation result in much more pronounced vertical turbulent pollutant flux near and above the roof level (Fig. 8c). In contrast to the heating simulation in which small-scale vortices more prominently contribute to upward turbulent pollutant flux near the roof level, the turbulent pollutant fluxes due to small-scale vortices near and well above the roof level are comparable with each other in the no-heating simulation.

How the contributions of the three bands to vertical turbulent pollutant flux vary depending on the location between buildings is examined for the heating and no-heating simulations (Fig. 9). At $z/H = 0.0625$, the contributions of the L and M bands to vertical turbulent pollutant flux are distinctly larger in the downwind regions of street canyons than in the upwind regions and their differences between the upwind and downwind regions are weakened by the surface heating. The contribution of the H band to vertical turbulent pollutant flux is also higher in the downwind regions in both heating and no-heating simulations,

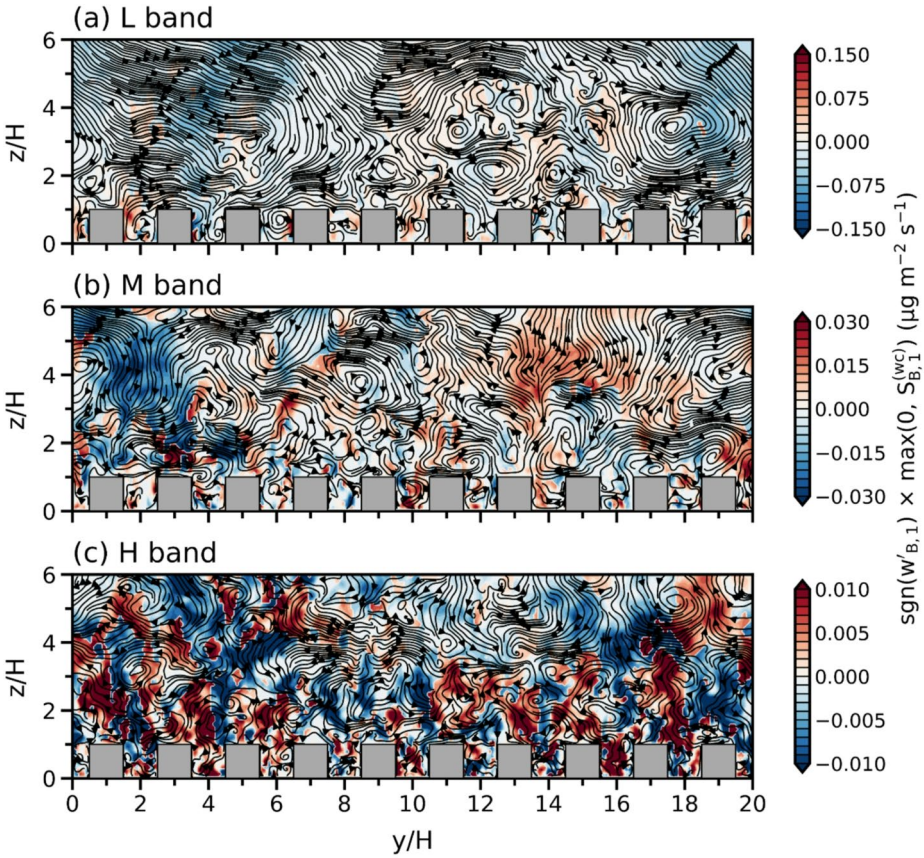

Fig. 8 Same as Fig. 7 except for the no-heating simulation

while its dependence on the location between buildings is relatively weak. At the roof level, the contributions of all three bands are overall larger in the downwind regions than in the upwind regions regardless of the surface heating. The surface heating weakens these differences between the two regions, as found near the pedestrian level.

3.4 Interactions among different scales of TCSs

To assess how large-scale coherent structures influence the intensities of small-scale coherent structures and their associated pollutant dispersion, the AM analysis is conducted in this subsection. Figure 10 presents the vertical profiles of AM coefficients of L band velocities with H band turbulent momentum and pollutant fluxes in the heating and no-heating simulations. In both simulations, the magnitudes of AM coefficient are small within the canyon while they are relatively large above the roof level (Fig. 10). For the modulations of the H band vertical momentum and pollutant fluxes by the L band streamwise velocity above the roof level, the AM coefficients show negative values (Fig. 10a, b). This suggests that in both simulations, small-scale vertical momentum and pollutant fluxes are enhanced in large-scale low-speed regions while they are suppressed in large-scale high-speed regions. For the modulations by the L band vertical velocity above the roof level, the AM coefficients show positive values (Fig. 10c, d). This suggests that small-scale vertical momentum and pollutant fluxes are enhanced in large-scale upward-motion regions while they are suppressed in large-scale downward-motion regions in both simulations. The large negative (positive) AM coefficients for large-scale streamwise (vertical) velocity above the roof level are also found in Salesky and Anderson [25]. The AM coefficients generally exhibit larger magnitudes in the heating simulation than in the no-heating simulation above the roof level. Thus, the surface heating acts to enhance the influence of large-scale coherent structures on small-scale turbulent momentum and pollutant fluxes above the roof level.

Figure 11 shows the spatial distributions of AM coefficient between the L band vertical velocity and H band vertical turbulent pollutant flux at $x/H=9$ in the heating and no-heating simulations. In the regions with positive (negative) AM coefficient, small-scale vertical turbulent flux is enhanced (suppressed) in large-scale upward motions and is suppressed (enhanced) in large-scale downward motions. Compared with the no-heating simulation, in

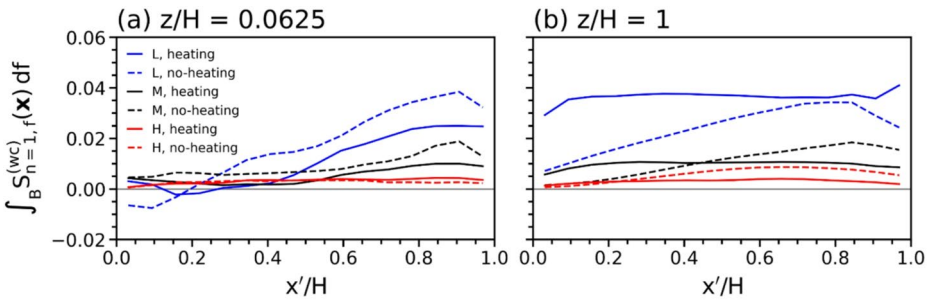

Fig. 9 Spectral proper orthogonal decomposition cospectra between vertical velocity and pollutant concentration for the first mode at $\mathbf{a}z/H=0.0625$ and $\mathbf{b}z/H=1$ during $t=2$–3 h in the heating (solid) and no-heating (dashed) simulations. The cospectra are averaged over the 100 lines between buildings with their y-locations being at the y-locations of the building centers within the domain. The blue, black, and red lines indicate the low-frequency band (L band), middle-frequency band (M band), and high-frequency band (H band), respectively

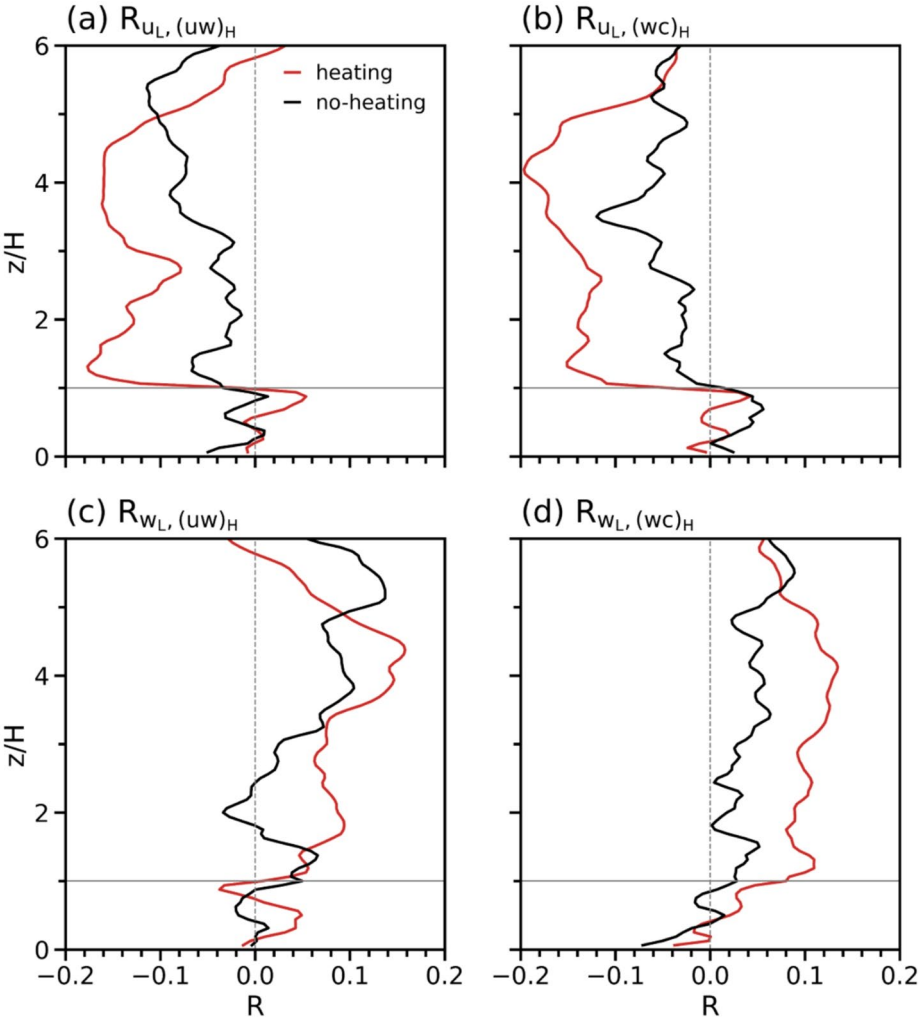

Fig. 10 Horizontally averaged vertical profiles of amplitude modulation coefficients of L band streamwise velocity (u_{L}) with H band vertical turbulent **a** momentum ($(uw)_{\mathrm{H}}$) and **b** pollutant ($(wc)_{\mathrm{H}}$) fluxes and those of L band vertical velocity (w_{L}) with H band vertical turbulent **c** momentum ($(uw)_{\mathrm{H}}$) and **d** pollutant ($(wc)_{\mathrm{H}}$) fluxes at $x/H=9$ in the heating (red) and no-heating (black) simulations

the heating simulation, the regions of positive AM coefficient are more well-organized in the spanwise and vertical directions (Fig. 11a, b). Particularly, at $y/H\sim 11$–13, the strong positive AM coefficient region with the value >0.3 extends from $z/H\sim 1$ to $z/H\sim 4$ in the heating simulation. For each sign of the L band vertical velocity at $z/H=1$–3, the conditionally averaged AM coefficient is calculated and reveals that the positive AM coefficient regions are mainly contributed by large-scale upward motions rather than downward motions in both simulations. This result suggests that the modulation of small-scale turbulent fluxes is mainly driven by large-scale upward motions regardless of the surface heating and the

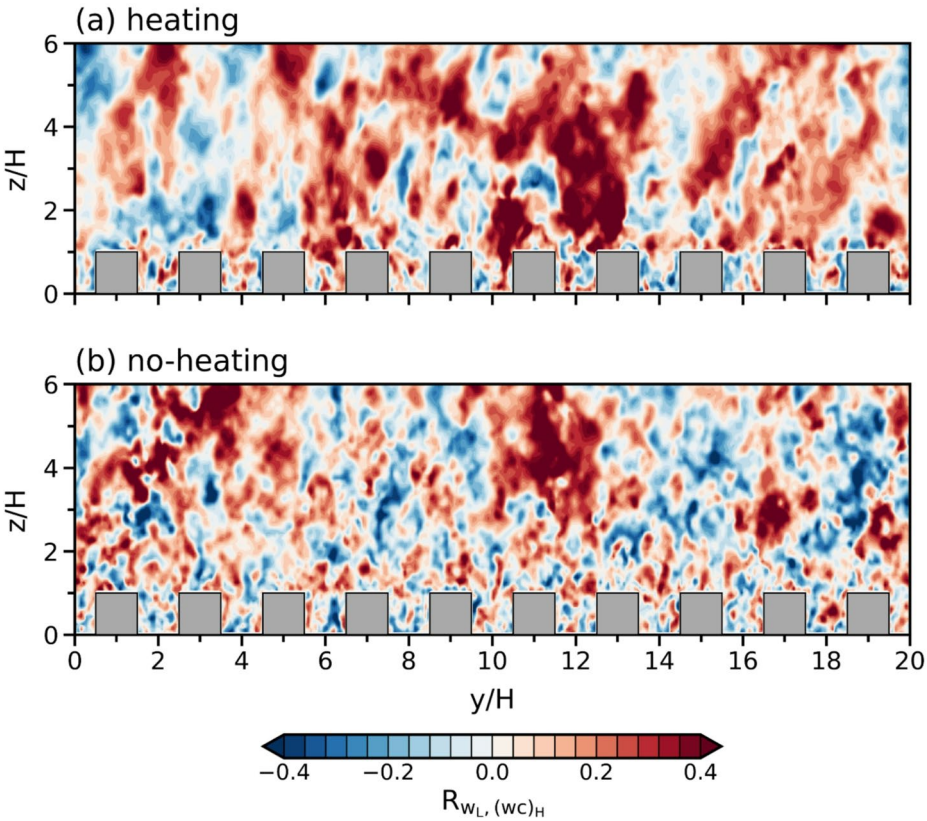

Fig. 11 Fields of amplitude modulation coefficient between L band vertical velocity and H band vertical turbulent pollutant flux at $x/H=9$ in the **a** heating and **b** no-heating simulations

enhancement of the modulation due to the surface heating is primarily associated with the strengthened large-scale upward motions.

We also evaluate the AM coefficients between the L and M bands and between the M and H bands in the heating and no-heating simulations. The AM coefficients between the L and M bands exhibit the vertical structures and spatial distribution (not shown) that are broadly similar to those found between the L and H bands (Figs. 10 and 11), with the AM being pronounced above the roof level and its magnitude being larger in the heating simulation than in the no-heating simulation. This result indicates that large-scale coherent structures also modulate the amplitude of mid-scale turbulent fluxes. Meanwhile, the AM coefficients between the M and H bands are negligible (not shown), implying that the small-scale turbulent flux is mainly modulated by large-scale coherent structures rather than mid-scale coherent structures.

In summary, the SPOD and AM analysis collectively reveal that the surface heating preferentially enhances large-scale coherent structures, increasing their contributions to the TKE and turbulent pollutant flux and enhancing their influence on small-scale coherent structures. The combined use of the two methods enables not only examining the heating impact on different scales of TCSs but also elucidating the heating impact on the interac-

tions among them. Applying an appropriate modal decomposition method together with a scale-interaction analysis has strong advantages in examining scale-dependent TCSs and their interactions in detail.

4 Summary and conclusions

In this study, LESs of an idealized building array with and without surface heating are performed to examine the modification of different scales of TCSs due to surface heating and its influence on turbulent pollutant flux. The SPOD is conducted to decompose turbulent flows into different scales of TCSs, and the AM analysis is conducted to examine the interactions among TCSs with different scales. Compared with the no-heating simulation, in the heating simulation, low-speed streaks are more pronounced and vertically well-developed and high-speed regions are less organized into streaks. The contributions of large-scale coherent structures to the TKE and turbulent pollutant flux at the roof level are considerably larger in the heating simulation than in the no-heating simulation. The amplitude modulation of small-scale turbulent fluxes by large-scale motions in the heating simulation is stronger than that in the no-heating simulation as the large-scale upward motions are strengthened due to the surface heating.

The study employs the SPOD to examine TCSs with different scales. Although the SPOD well captures optimally energetic spatiotemporal patterns that are statistically dominant, it does not guarantee the dynamical coherence of these patterns. Future investigation using an analysis method based on dynamics is needed to corroborate the TCS characteristics and interactions identified in this study. Including the present study, a number of studies use the frequency to resolve different sizes of turbulent flows. Applying a method to decompose turbulent flows into different scales of TCSs directly based on their sizes could be beneficial for better investigating the scale-dependent characteristics of TCSs. The AM analysis in this study reveals that the amplitudes of small-scale turbulent fluxes are correlated with large-scale motions, suggesting the modulation of small-scale coherent structures by large-scale coherent structures. While the statistical coupling between coherent structures with different scales can be diagnosed by the AM analysis, the causality of this statistical coupling cannot be verified through the statistical approach. How much the large-scale (small-scale) coherent structures dynamically modulate small-scale (large-scale) coherent structures needs to be investigated. In addition, further examinations considering a wide range of building configurations, heating intensities, and flow regimes can help to better understand TCSs with different scales and their roles in pollutant transport in urban areas, as well as to supplement the findings of this study summarized above.

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Author contributions Conceptualization: Jong-Jin Baik; Methodology, data curation, and visualization: Dong-Hwi Kim; Formal analysis and investigation: Dong-Hwi Kim, Seong-Ho Hong, Kyeongjoo Park, Jong-Jin Baik; Writing – original draft: Dong-Hwi Kim, Seong-Ho Hong, Kyeongjoo Park; Writing – review & editing: Beom-Soon Han, Jong-Jin Baik; Supervision: Beom-Soon Han, Jong-Jin Baik.

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Data availability The data that support the findings of this study are available from the corresponding authors upon reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

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